Update – Sep 18

JC Vaught

Fellowships

- DOD SMART – Dec 5
  - 1 in 12 approval; avg GPA 3.7; up to $46k/yr stipend; 5yr funding
  - https://www.smartscholarship.org/apply
- NDSEG(National Defense) – Nov 15
  - 1 in 50 approval; $40.8k stipend; 3 yr funding; Merit-based
  - https://applyndseg.sysplus.com/Login
- NSF GRFP – Nov 13
  - 1 in 5 approval; $37k stipend; 3 yr funding; Merit-based
  - https://www.research.gov/grfp/Login.do
- DOE CGSF – Jan 16??? Not available
  - 1 in 20 approval; $45k stipend; 4yr* funding; Merit-based

- Hertz fellowship – Oct 31
  - 1 in 40 approval; ~$35k stipend; 5yr funding; Merit-based(but not really)
  - https://fellowship.hertzfoundation.org/apply/

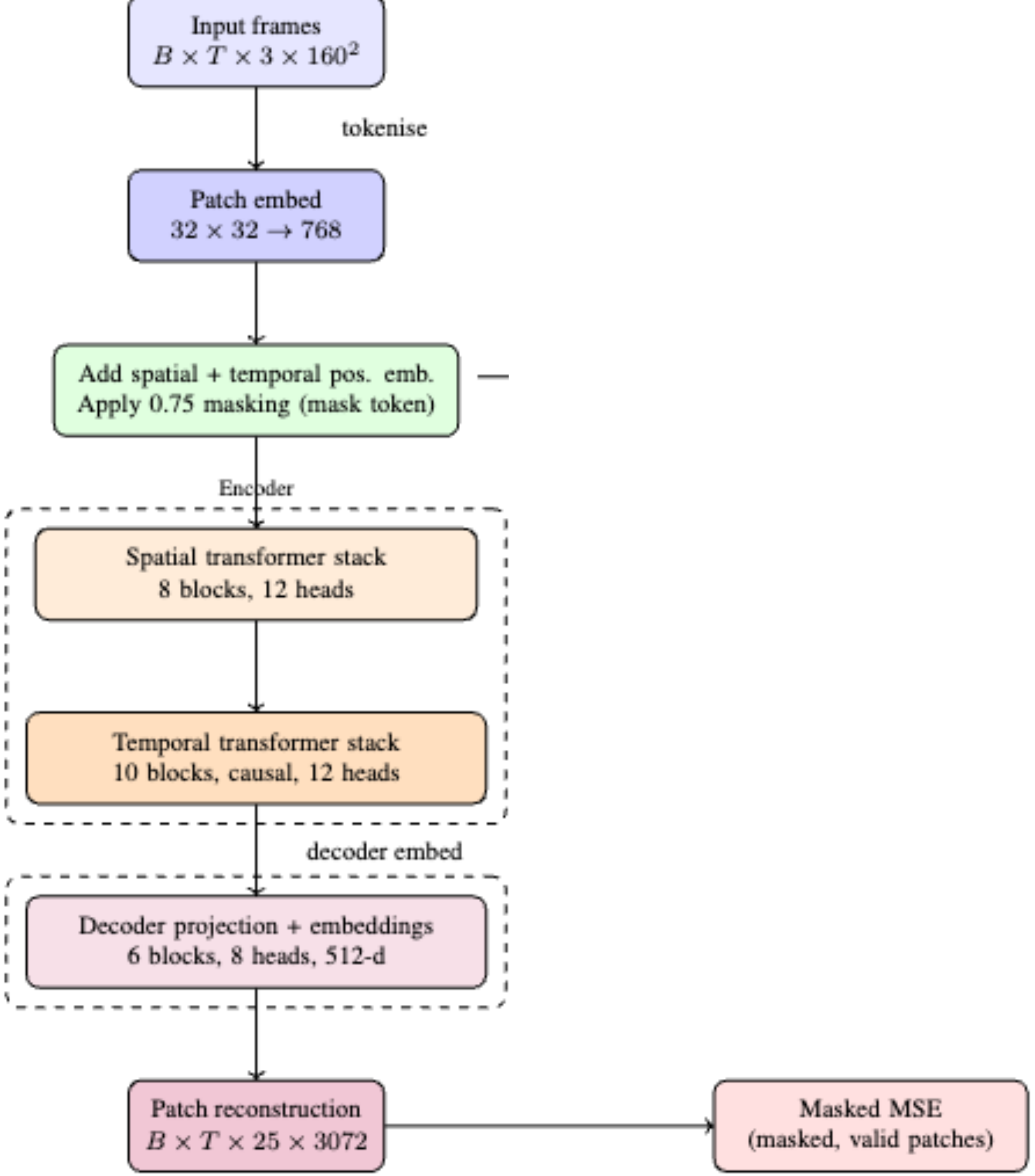
- AIAA – Jan 31
  - ~1 in 33 odds
  - $1k to $10k
- ASME – Feb 26
  - ~1 in 10 odds
  - $0.5k to $10k

Progress

- Last week
  - Attempted Temporal-MAE for gain prediction
    - Full Frame masking
- This week
  - Rewrote to add variable context design
    - Adds more frames as they become available(i.e. real time)
- Now/Next week
  - Adding Full-Resolution capabilities
  - Sharding model across GPUs(running into OOM issues)

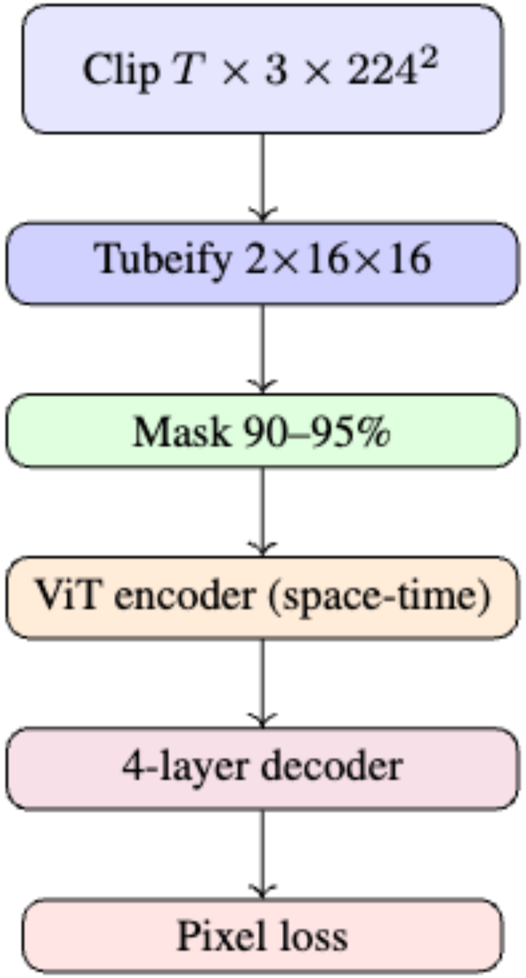
Architecture

- Similar to



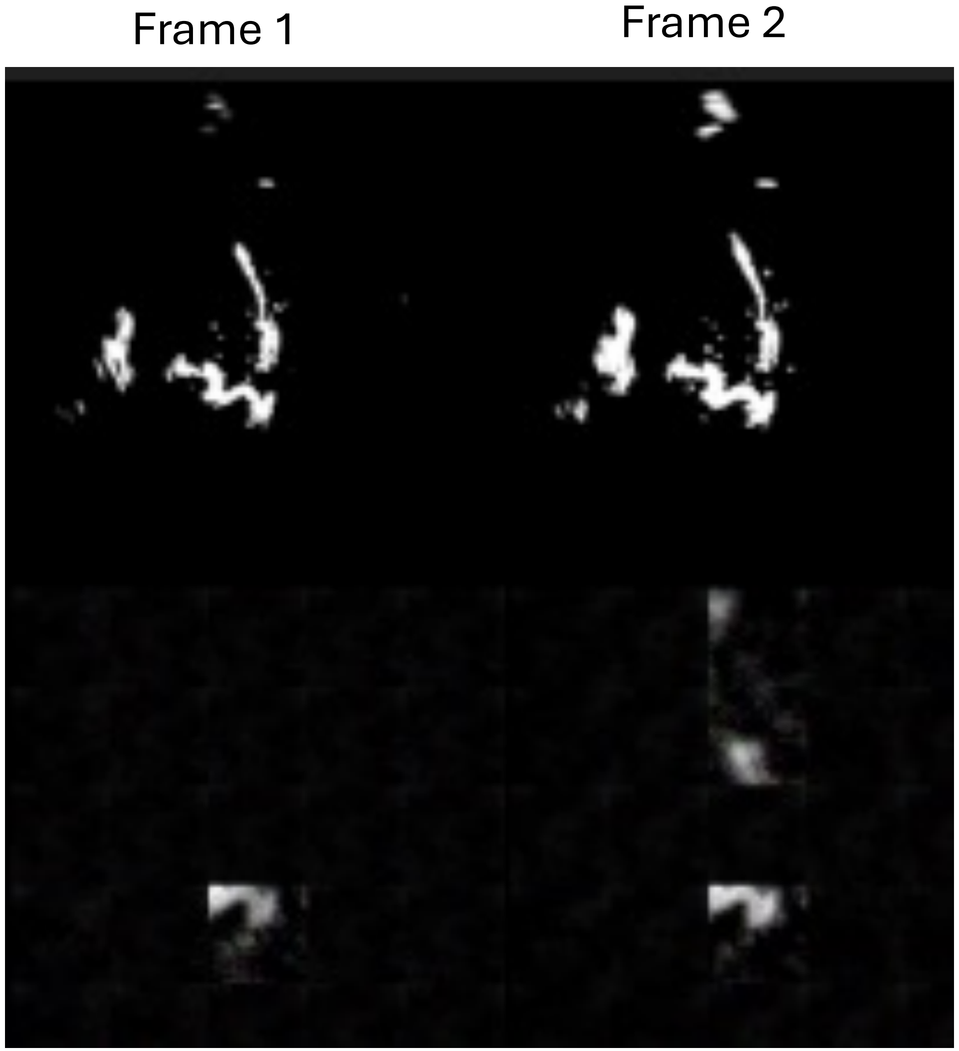
Architecture VideoMAEv1

- What my system is based on basically

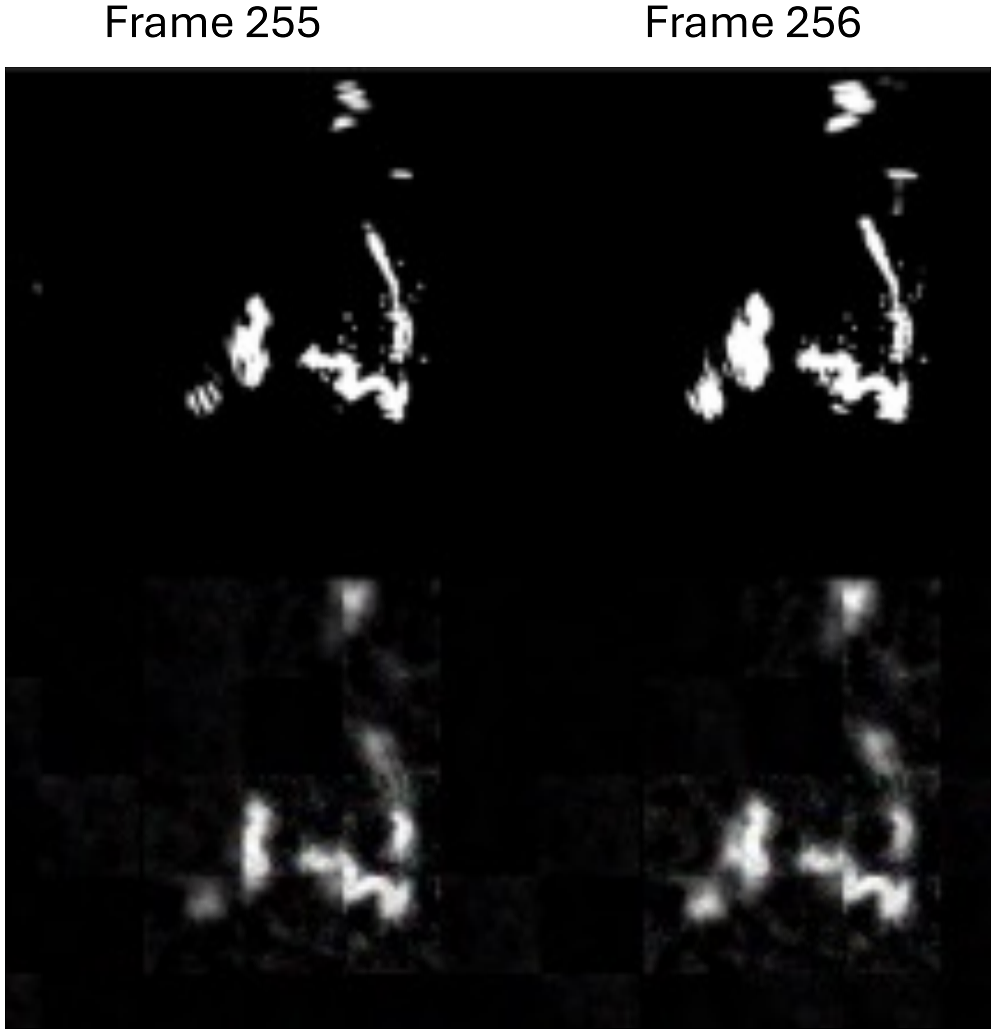


(a) VideoMAE[9]

Variable Context



Predicted output

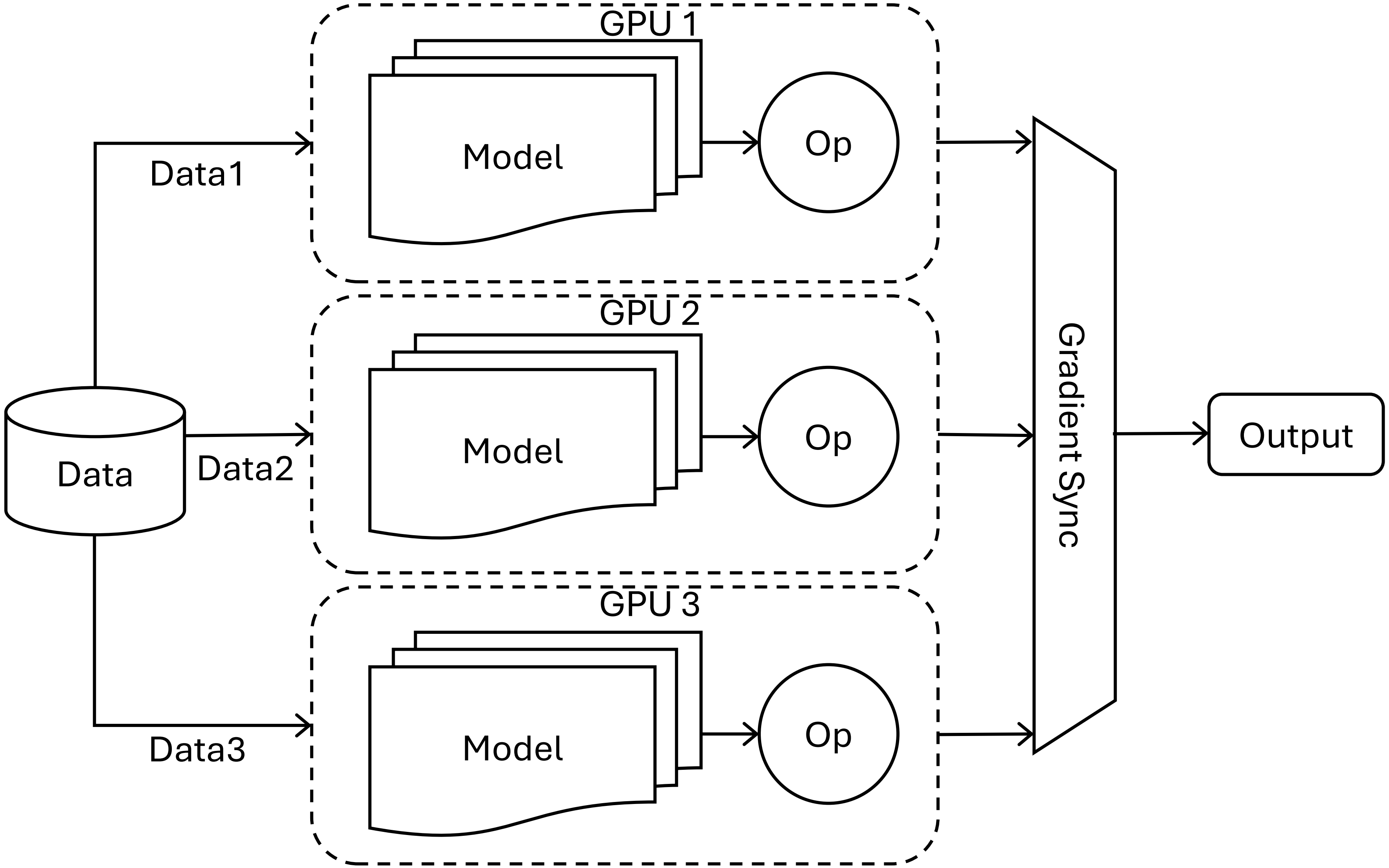


Predicted output

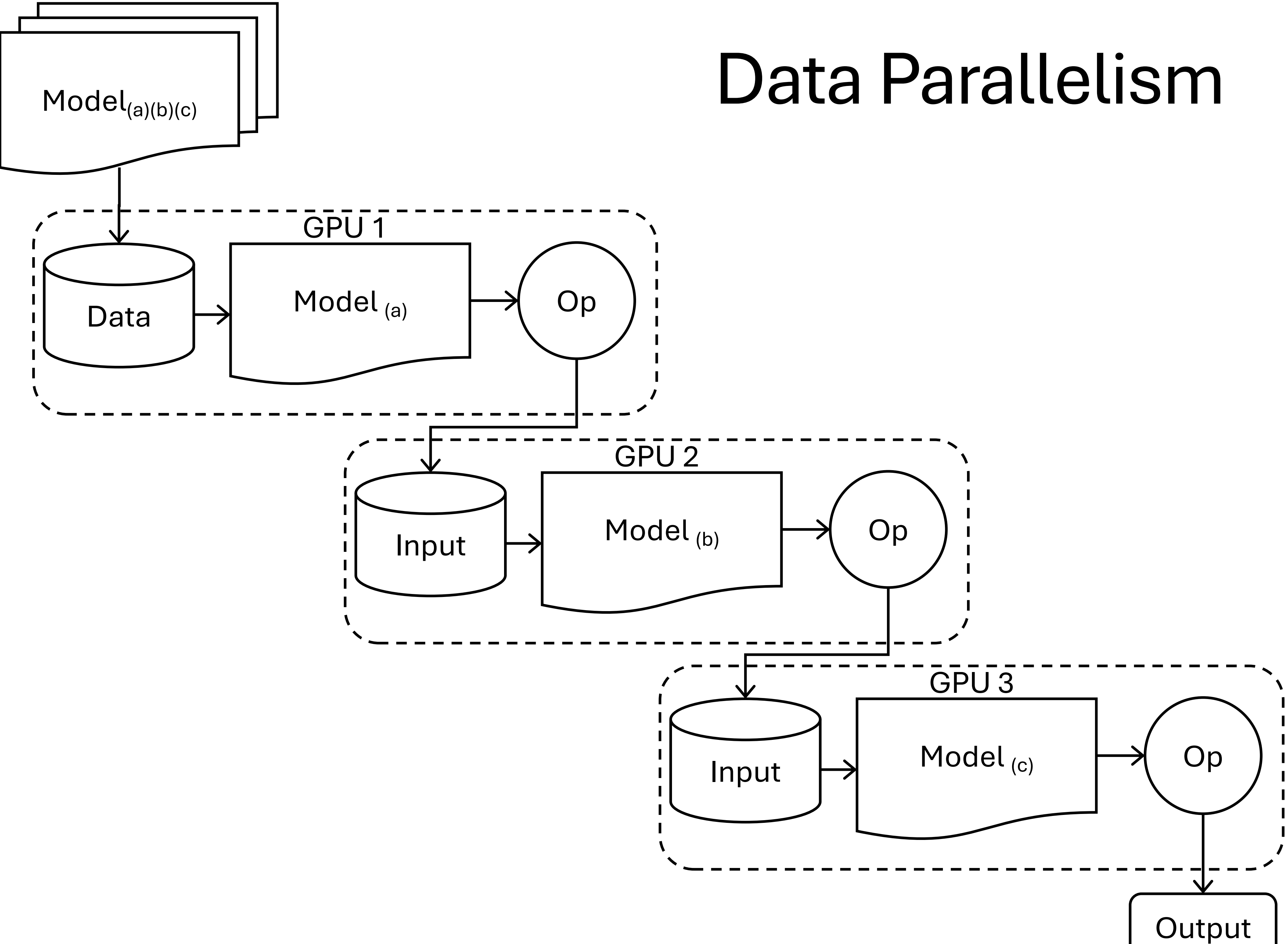
Splitting Load up between GPUs

- **4 Options**
  - Data Parallelism
  - Model Parallelism
  - Pipeline Parallelism
  - Tensor Parallelism

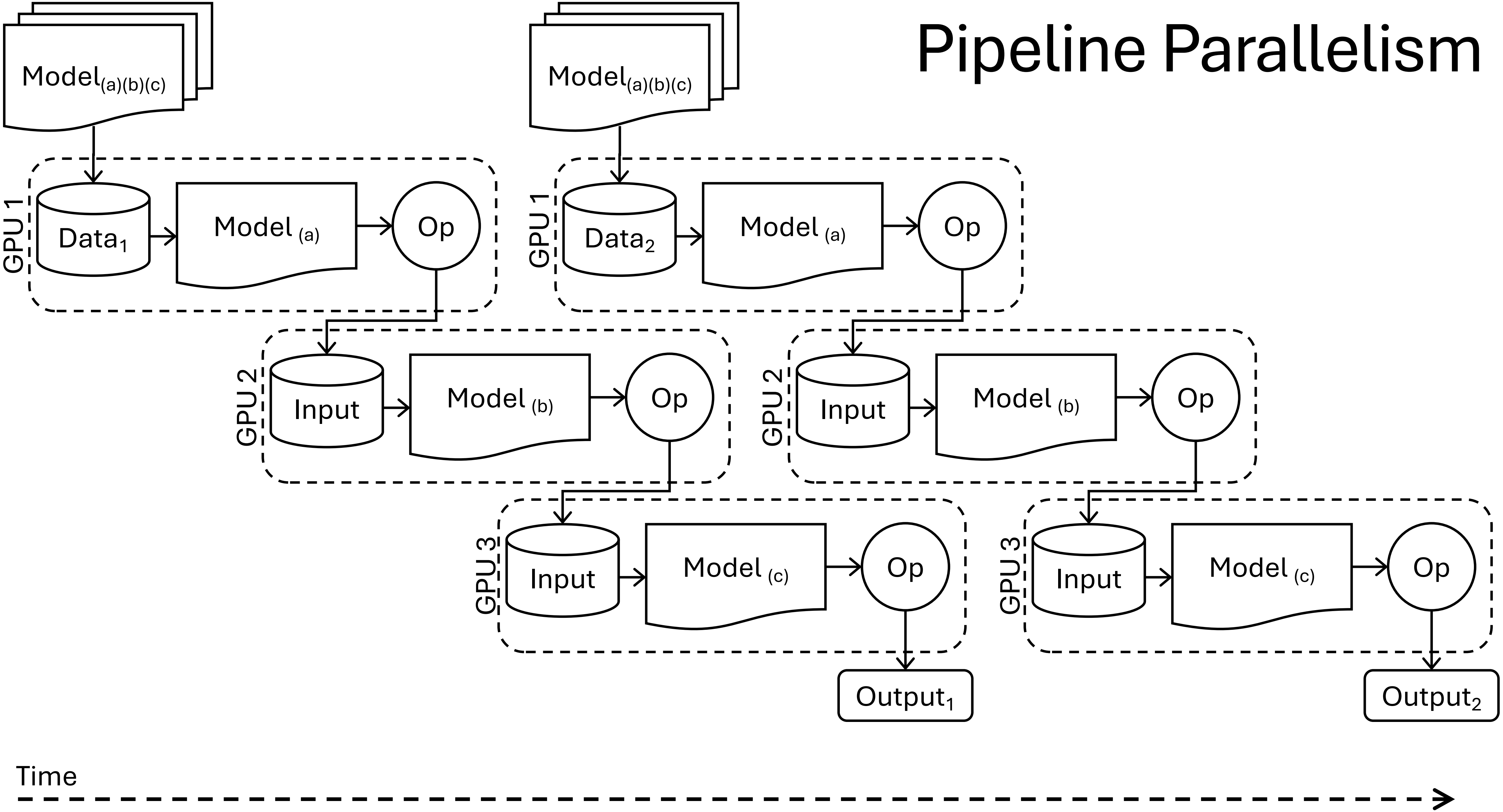
Data Parallelism

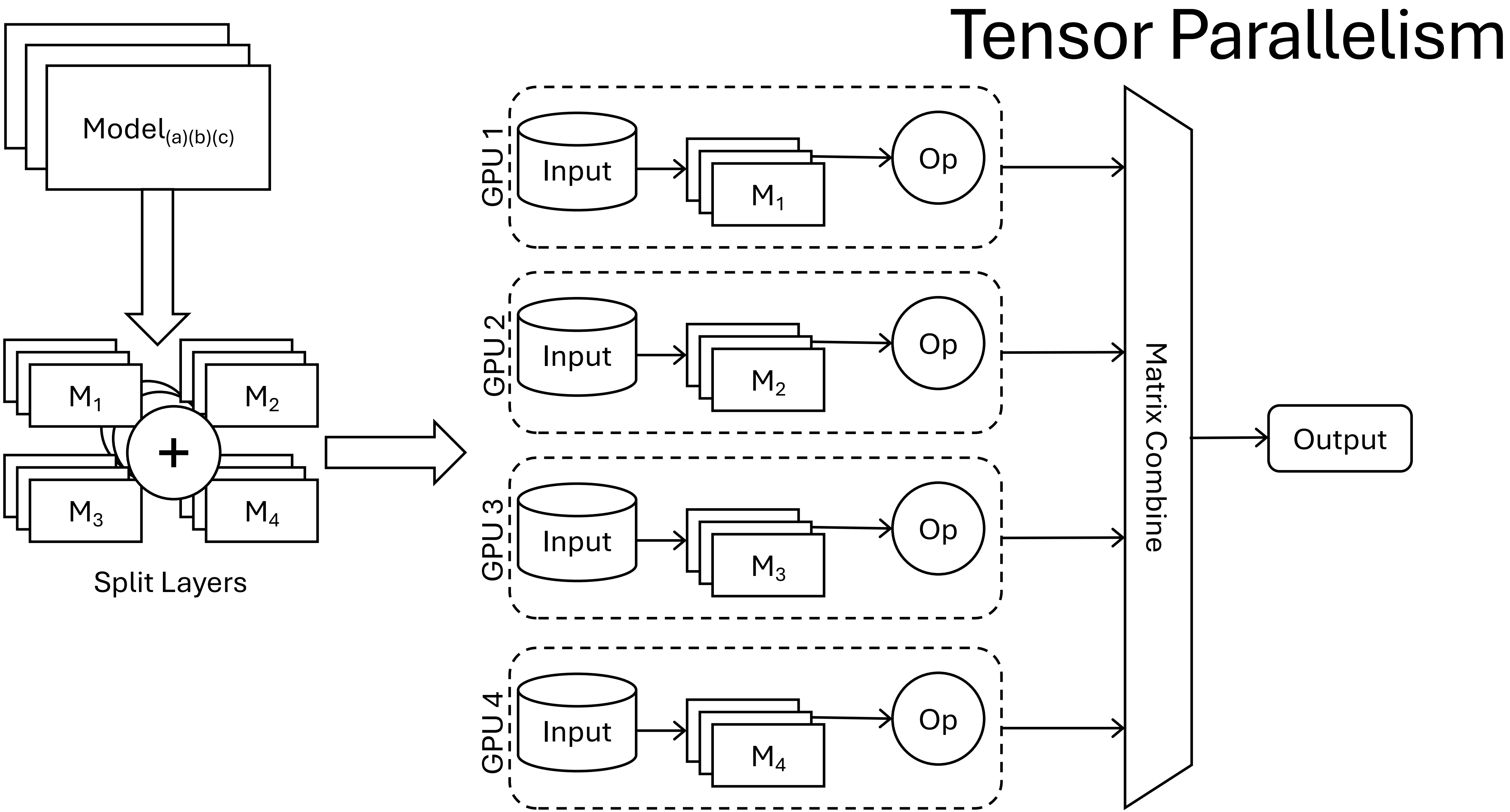


Data Parallelism



Pipeline Parallelism





Tensor Parallelism
Model(a)(b)(c)
M1
M2
M3
M4
+
Split Layers
GPU 1
Input
M1
Op
GPU 2
Input
M2
Op
GPU 3
Input
M3
Op
GPU 4
Input
M4
Op
Matrix Combine
Output